D-001 · ART DIRECTION DELIVERABLE

PANDEMONIUM
ART BIBLE

The single visual direction for every Pandemonium asset: a readable, stylized RTS look built silhouette-first. This bible fixes the palette, material language, lighting and scale rules for units, buildings and the Scavenger and Ironclad concept armies.

E-Vex · pandemonium-assets
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SILHOUETTE FIRST · TEAM COLOR SECOND · DETAIL LAST

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1. Direction and Principles

Pandemonium renders a battlefield from a high-angle real-time strategy camera, and every asset must survive that viewing condition. The direction is a readable stylized look, not a realistic one: low-poly geometry, matte surfaces and bold shapes that resolve instantly at gameplay zoom. The current tactical_field textures are an interim look and must not be treated as a style target. Where a trade-off appears, readability always wins over fidelity, and fidelity always wins over decoration.

Every visual decision is ranked by a fixed priority stack. First, the silhouette: the shape alone must identify the unit's role before any color or detail is visible. Second, the team color: one flat, bright zone communicates ownership at a glance across the whole roster. Third, detail: surface texture, greebles and small parts come last and only where camera time actually exists. Detail that cannot be perceived from the RTS camera is budget spent on nothing, and it actively hurts the silhouette when it adds spikes and noise.

The camera looks down at roughly 55 degrees with a near-orthographic feel. That means top surfaces are prime real estate: roofs, shoulder plates and headgear carry the light, the team color and the identifying shapes. Fronts and sides are secondary; undersides are almost never seen and stay unarticulated. Design every asset top-down first, then check the profile, then the front.

The readability budget in practice:

- Every unit and building must be identified from its silhouette alone in a 48 x 48 pixel thumbnail (see the role chart in chapter 5).
- One dominant shape per asset: a wedge, a block, a disc, a tower. Secondary protrusions never compete with the dominant shape.
- Team color is one flat zone family per asset, always in the same location family so the eye knows where to look.
- Surfaces are matte with 3-value shading. Nothing gleams, nothing reflects the sky, nothing blooms.
- Detail density is capped by the 48 px test: if a feature vanishes at 48 px, it belongs to the D-002 model pass, not to the gameplay read.

2. Color System

The whole game draws from one fixed palette: 12 base colors for all factions and materials, plus 8 team colors reserved strictly for player ownership. The base palette is a military-industrial family: desaturated grays for structures, warm tans and browns for fabrics and wood, and three controlled accents. Because bodies stay desaturated, the saturated team colors pop without any glow effects, and a recolor pass never has to touch the underlying material work.



Figure 2-1: the 12-color base palette with usage notes, and the 8 team colors.

Area discipline follows a simple ratio. Roughly 60 percent of any asset is body neutral (gunmetal, slate, soot), 25 percent is secondary material (canvas, concrete, tires), 10 to 15 percent is team color, and accents stay under 5 percent. When an asset feels wrong, measure the team-color area first: too little and ownership is invisible, too much and the asset reads as a toy.

Team colors live in a narrow value band so no player is brighter or darker than another. Each asset carries exactly one primary team-color zone on its upper mass (shoulders, roof, mantlet) plus at most one secondary trim. Team color never lands on thin parts that alias away at distance - antennas, barrels, thin struts - and never appears as scattered specks. For players with color-vision deficiency, team hues alone are not sufficient; the fallback is per-team emblems and trim patterns, to be finalized with the client UI in D-002.

The two signal accents have hard rules. Signal yellow is a hazard color and always appears striped with soot black, which keeps it distinct from team yellow. Glass cyan is the only emissive material and covers

at most 5 percent of an asset: windows, energy cells, screens. It marks function, not decoration.

Kind	Primary team-color zone
worker	hard hat + backpack shell
rifleman	torso vest + helmet band
raider	engine cowl
guardian	shoulder mantlet plates
command center	roof disc + door frame
barracks	gable roof panels
supply depot	silo dome caps
turret	mantlet ring
ore node	none (neutral)

Table 2-1: where the team-color zone sits on each kind.

3. Material Language

Materials are defined by finish, not by texture resolution. Everything is matte; the differences come from value steps and small color temperature shifts, which survive compression, recolors and the long camera distance. The table below is the complete material vocabulary for every faction - Scavengers simply weight it toward rust and canvas, Ironclad toward gunmetal and bone white.

Material	Finish	Wear rules	Notes
Painted team plate	flat, gloss 0-5%	no chipping inside team zones	the ownership surface; keep clean so recolors stay trivial
Bare construction metal	gunmetal, matte	edge wear only on handling edges	primary structure color
Rubber and tires	near-black, zero gloss	none	wheels, tracks, grips
Canvas and tarp	tan, zero gloss	big soft folds only	low-poly friendly; Scavenger signature
Rust and scrap	rust accent, matte	patchy plate borders	improvised look; Scavenger buildings and vehicles
Glass and energy	emissive cyan	none	max 5% of asset area; function beacons only

Material	Finish	Wear rules	Notes
Concrete	flat, faint tint variance	hairline cracks at most	building foundations and walls
Wood and crate	leather brown	plank gaps	supply visuals

Table 3-1: the complete material vocabulary.

The forbidden list is as important as the vocabulary: no chrome or mirror finishes, no specular bloom or HDR glints, no procedural grime noise at gameplay scale, no alpha-decal clutter, and no glossy team paint. Any effect that makes one asset shinier than its neighbors breaks the flat, readable field the whole roster depends on.

4. Lighting

One key light, baked once, identical for every asset and faction. The key comes from the north-west at azimuth 315 degrees and 60 degrees elevation, so top faces facing NW catch the light and SE faces fall into shadow. Filling is a cool sky ambient at roughly 35 percent, plus a faint warm ground bounce on downward-facing surfaces. Because the light is fixed, players learn shape-from-shading once: a lit top plane means the same thing on every unit, and no asset ever fights its neighbor's lighting direction.

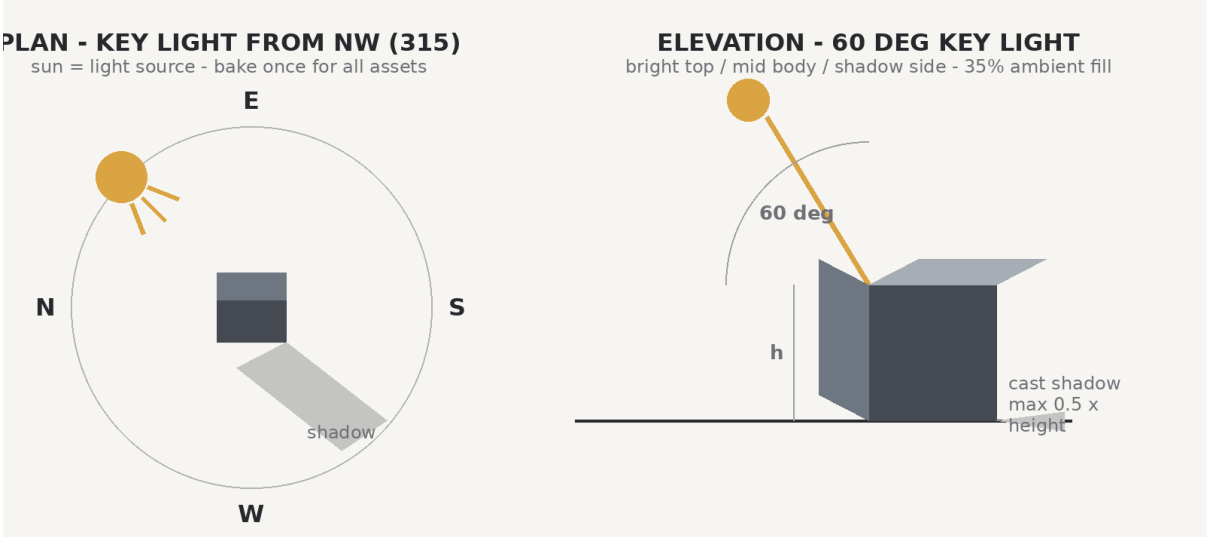


Figure 4-1: key light from the NW in plan, and the 60-degree elevation with 3-value shading.

Shading on any asset uses three values: a bright top plane, a mid body and a shadow side. Cast shadows are soft, short and never longer than half the asset's height, keeping the ground readable when a base fills with buildings. All shading is baked into textures and vertex color; the client applies no dynamic per-asset lighting that could break this convention.

- Key light: azimuth 315 (from NW), elevation 60 degrees, warm-neutral white.

- Ambient fill: cool sky tint, about 35 percent.
- Ground bounce: warm 10 percent, only on downward faces.
- Cast shadow: soft, max 0.5 x asset height, always toward SE.
- No rim lights, no colored spotlights, no per-faction lighting moods.

5. Silhouette Rules per Role

Each role owns a silhouette grammar - a shape language about mass distribution, protrusion count and one signature element. The grammar is what lets players parse a crowded screen without reading labels. If two roles drift toward the same profile, one of them is wrong and must be pushed apart.

Role	Mass language	Protrusions	Signature element
Worker	small, hunched, asymmetric	many small, harmless	oversized tool arm + backpack
Infantry	slim vertical figure	one: the rifle line	rifle held out horizontally
Fast raider	long, low horizontal wedge	few, swept back	pointed nose, exposed wheels
Heavy	tall and wide boulder	few and blunt	huge shoulder mantlet, small head
Defense	static pedestal + gun axis	one: the barrel	octagon base with long barrel
Headquarters	tallest landmark	vertical spire	antenna mast visible from anywhere
Production	long horizontal barn	one: the bay door	big door things emerge from
Supply	low chunky stack	repeated module bumps	paired silos and crates

Table 5-1: silhouette grammar per role.

The current eight kinds demonstrate the grammar. Each pair below shows the high-angle concept render next to its silhouette-only version; both must read as the same role. The renders use red as the demonstrative team color.

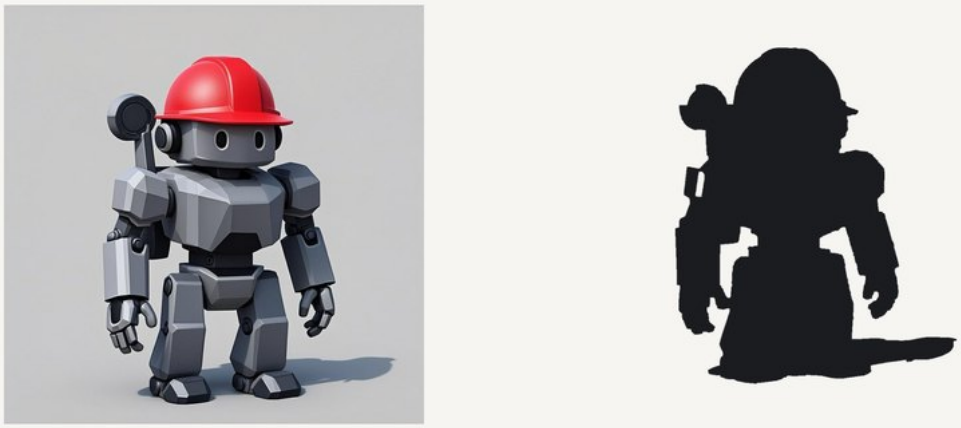


Figure 5-1: worker - high-angle render (left) and silhouette (right).

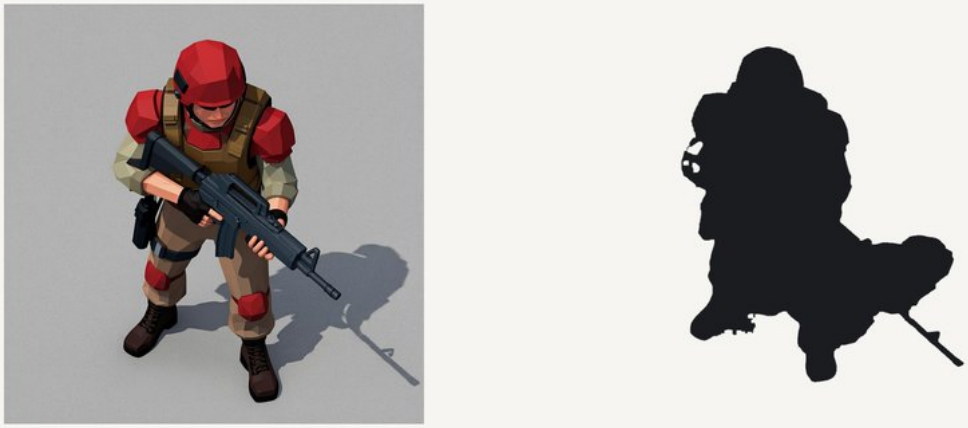


Figure 5-2: rifleman - high-angle render (left) and silhouette (right).

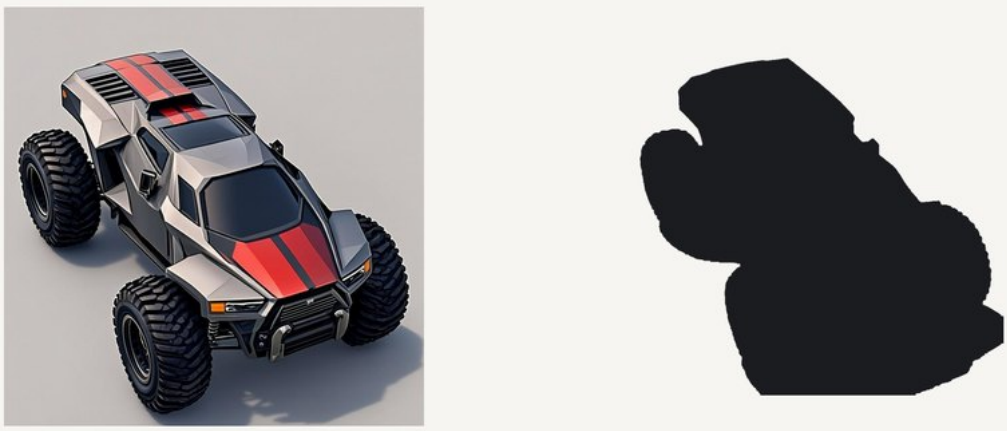


Figure 5-3: raider - high-angle render (left) and silhouette (right).





Figure 5-4: guardian - high-angle render (left) and silhouette (right).





Figure 5-5: command center - high-angle render (left) and silhouette (right).





Figure 5-6: barracks - high-angle render (left) and silhouette (right).



Figure 5-7: supply depot - high-angle render (left) and silhouette (right).

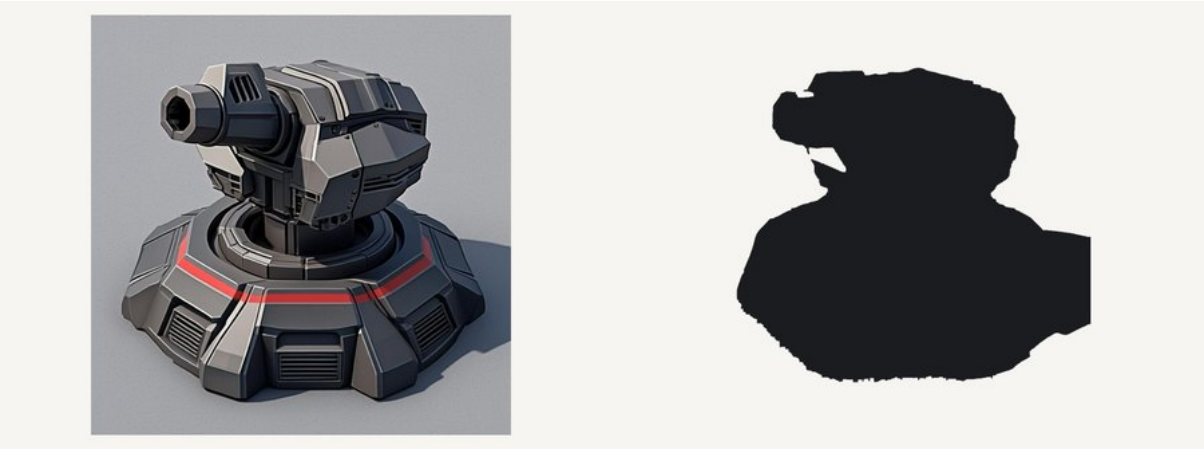


Figure 5-8: turret - high-angle render (left) and silhouette (right).

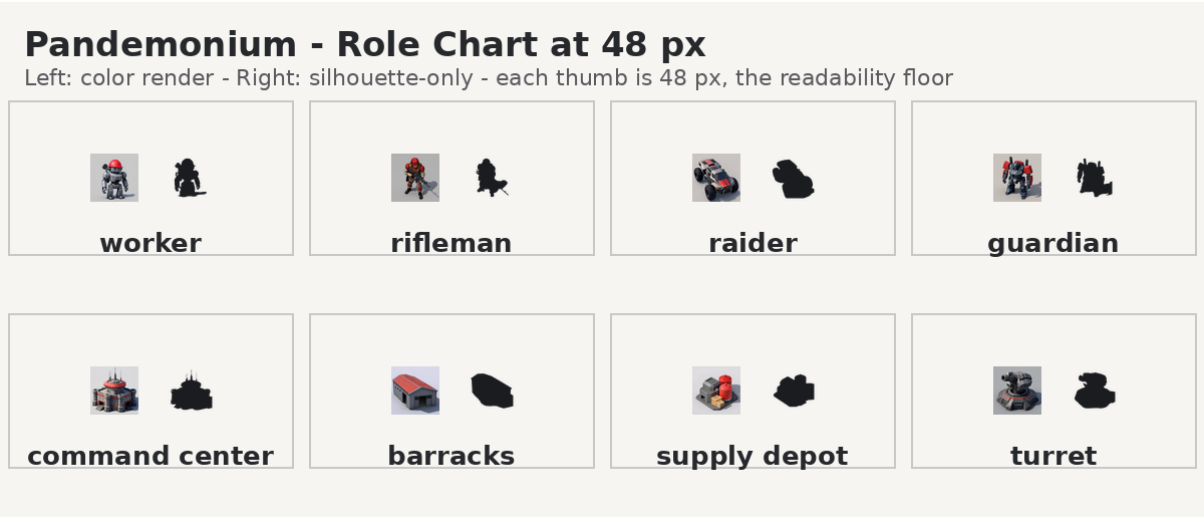


Figure 5-9: the 48 px readability test - every kind, color and silhouette.

The 48 px chart is the acceptance test for every future asset, including the faction armies in chapter 8. If a new unit cannot be named by a playtester from its 48 px silhouette alone, the silhouette is rejected

regardless of how good the full render looks.

6. Scale and Footprints

One tile equals 3 meters. Units are deliberately oversized, roughly one and a half times their real-world equivalents, because a realistic soldier would be invisible at RTS zoom. Widths are collision diameters; buildings occupy exact tile rectangles. Widths mirror the game's content data (content/entities, re-derived 2026-10-25: rifleman 0.7, raider 0.6; corrected in D-002). The command center is the landmark: the tallest object on the map, four by four tiles, visible from any fight.

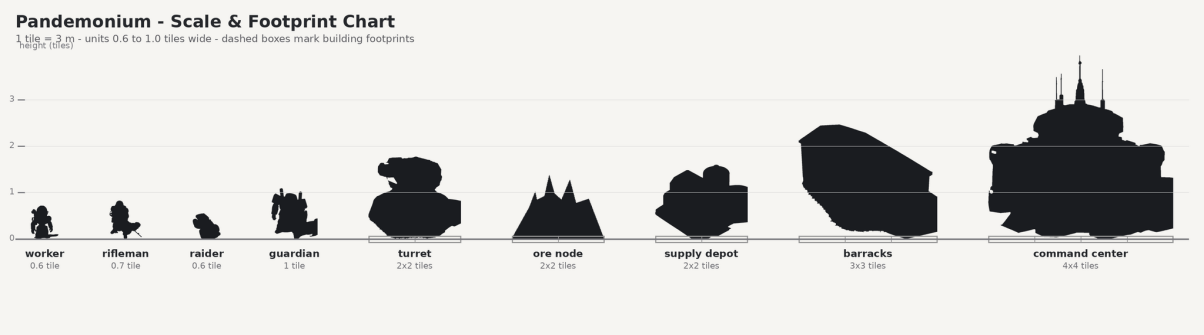


Figure 6-1: all kinds at true relative scale on the tile grid; dashed boxes mark building footprints.

Kind	Footprint	Size class	Height (tiles)
worker	0.6 tile wide	S	0.5
rifleman	0.7 tile wide	S	0.7
raider	0.6 tile wide	M	0.4
guardian	1 tile wide	L	1.2
command center	4x4 tiles	XL	3
barracks	3x3 tiles	L	1.2
supply depot	2x2 tiles	M	0.9
turret	2x2 tiles	M	1.5
ore node	2x2 tiles	M	0.8

Table 6-1: footprints and heights (1 tile = 3 m).

Heights follow a strict ladder: workers stay under half a tile, combat units cluster between half and one tile, the turret's barrel tip reaches one and a half tiles, and the command center tops three tiles. Anything taller than the command center is reserved for campaign landmarks, never for buildable assets, so the

ownership read of the skyline stays predictable.

7. Do and Don't

The four panels below compress the whole bible into one page of contrasts: silhouette clutter, team-color scattering, lighting chaos and samey mass. They are the fastest review tool - hold new concept work next to this page and the failure mode, if any, is usually obvious within seconds.

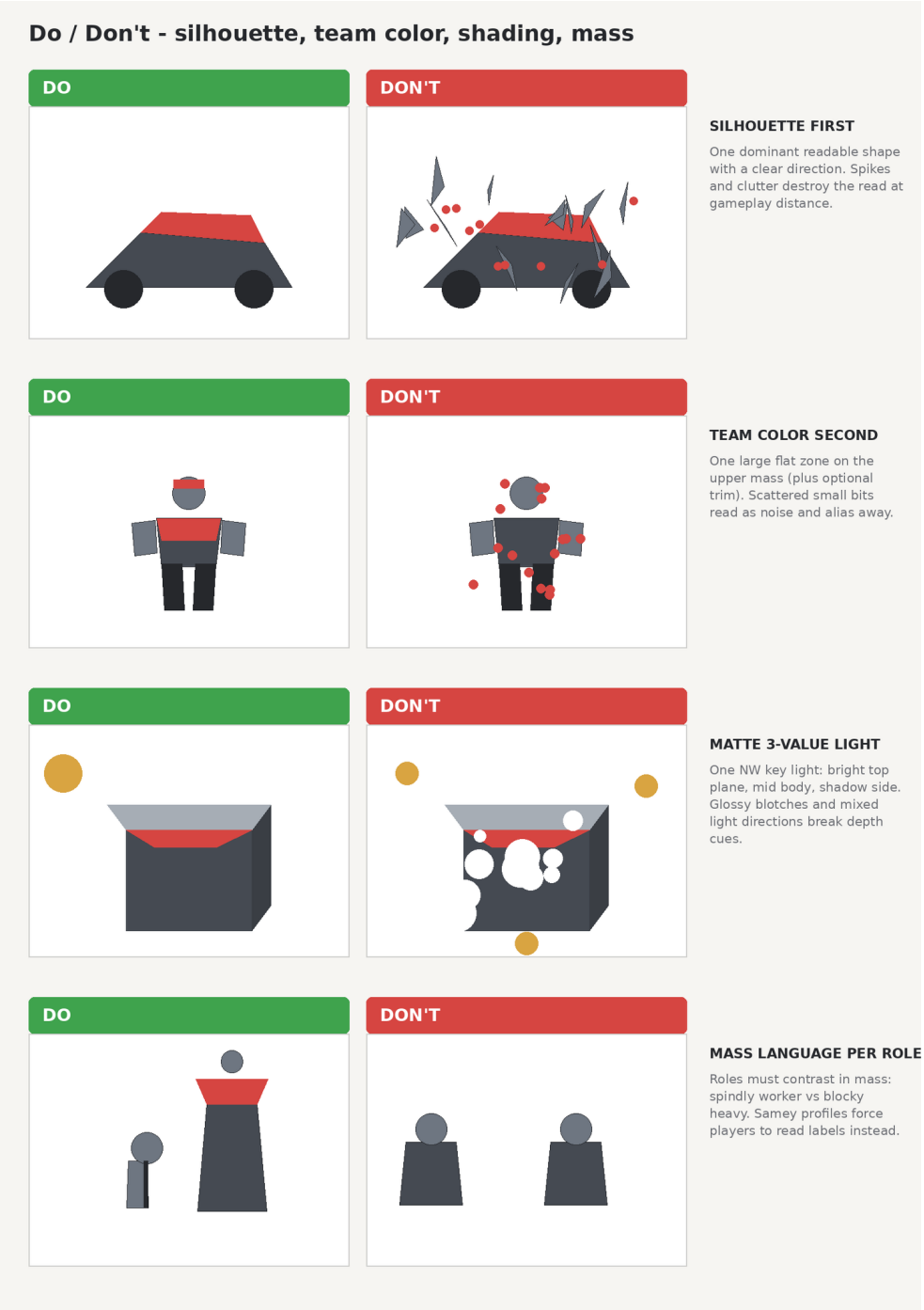


Figure 7-1: the four recurring failure modes, each paired with the correct treatment.

- Do keep one dominant shape with a clear direction; don't scatter equal-size protrusions that erase the read.
- Do place one large flat team zone on the upper mass; don't sprinkle small colored bits that alias into noise.
- Do shade in three values from the single NW key; don't mix light directions or add glossy blotches.
- Do exaggerate mass contrast between roles; don't let a worker and a heavy share a profile.
- Do desaturate bodies so team color pops; don't let faction materials compete with ownership color.

8. Faction Languages

Factions share the palette, lighting and readability rules, and separate through silhouette language and material weighting. The base army is the reference point: clean industrially manufactured shapes, symmetric, disciplined. Each concept army then bends exactly two dials - mass language and material mix - so the roster stays readable while the faction identity is unmistakable even in silhouette.

Scavengers are cheap, fast and improvised. Their silhouette language is jagged and asymmetric: mismatched plates at odd angles, exposed wheels, tarps and pipes. Materials weight toward rust, canvas tan and worn tires. Their buildings are visibly weak - thin scaffolding, stacked containers, tarp doors - telling every opponent where to hit. The lineup below is the faction readability check: seven entities, each namable at 48 px.

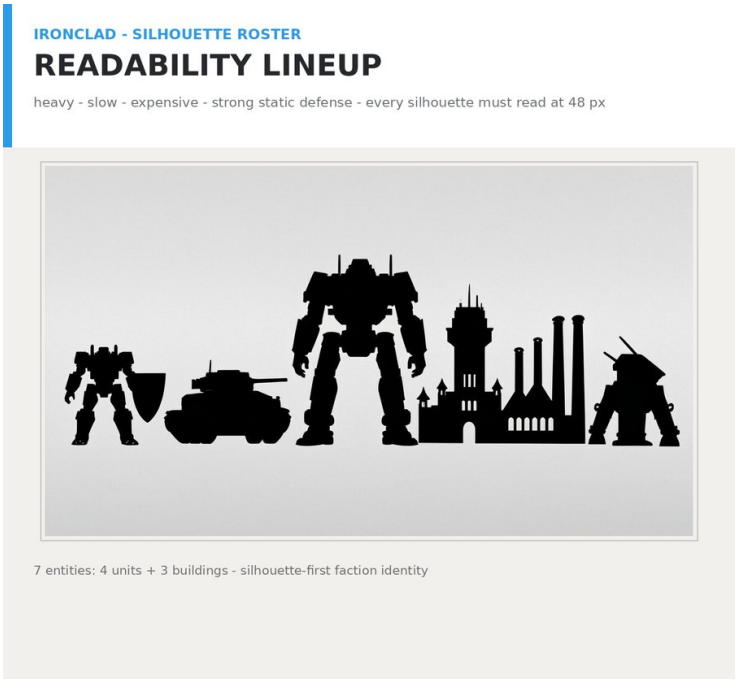


Figure 8-1: Scavengers silhouette roster (4 units + 3 buildings).

Ironclad are heavy, slow and expensive, with the strongest static defense in the game. Their language is the monolithic slab: thick symmetric riveted plating, fortress architecture, steam vents. Materials weight toward gunmetal and bone white with brass fittings. Buildings are bunkers - thick walls, star-fort angles, embedded guns. Where Scavengers look bolted together overnight, Ironclad look poured in one piece.

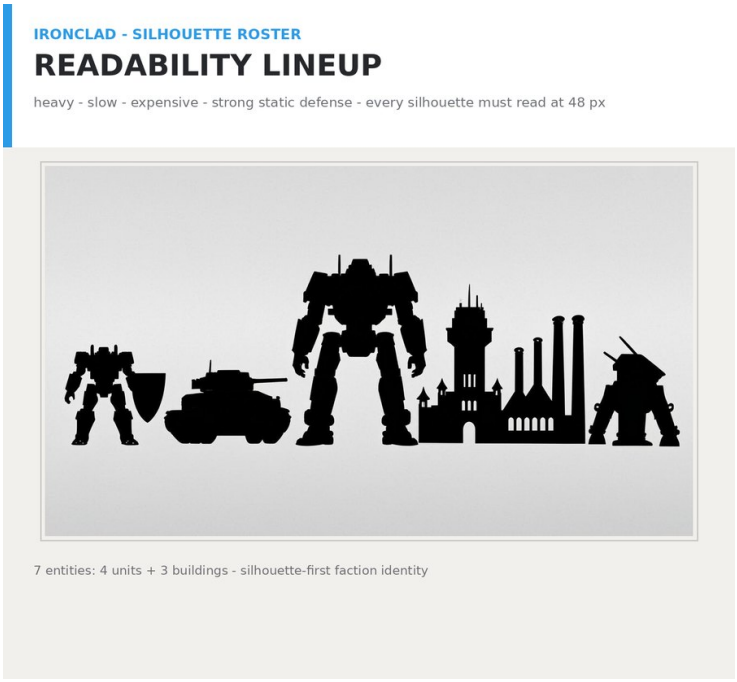


Figure 8-2: Ironclad silhouette roster (4 units + 3 buildings).

Full concept boards for both armies - front, side and top views with team-color zones marked and speed, mass and cost tiers - live in concepts/scavengers/ and concepts/ironclad/ in the repository. The rosters are: salvager, ratter, scrap_hound, ram, scrap_yard, garage, stash; and artificer, legionnaire, crawler, colossus, citadel, forge, bastion. These kind ids are proposals until engineering confirms them.

9. Delivery Conventions

Files are named in snake_case and must match the kind ids used by the client, exactly as the existing pack does: rifleman.png, never Rifleman_v2.png. No version suffixes, no capitals, no spaces. When a kind id changes, the file changes with it; there is no alias layer.

- Icons: 128 x 128 PNG with alpha (RGBA), transparent background, plus 512 px masters under icons/512/.
- Role renders: roles/.png as generated; silhouette versions roles/_silhouette.png with transparent background.

- Concept boards: concepts//.png, one board per entity, front/side/top views with team-color zones marked.
- Repository limits: PNGs max 2048 x 2048 and 64 MiB per file (client loader), and 25 MB per file for D-001 deliveries.
- Layered sources (.blend, .psd) are NOT committed yet; Git LFS will be set up first, and large sources go through the PM.
- The technical asset spec - model format, poly budget, axes and export scale - arrives in D-002 once the model pipeline exists.

The HUD icon set below is the current delivery: nine kind icons plus the three command icons attack_move, stop and guard, drawn as flat glyphs with a soot keyline so they stay readable on any terrain. Team red marks unit ownership; signal yellow marks commands.

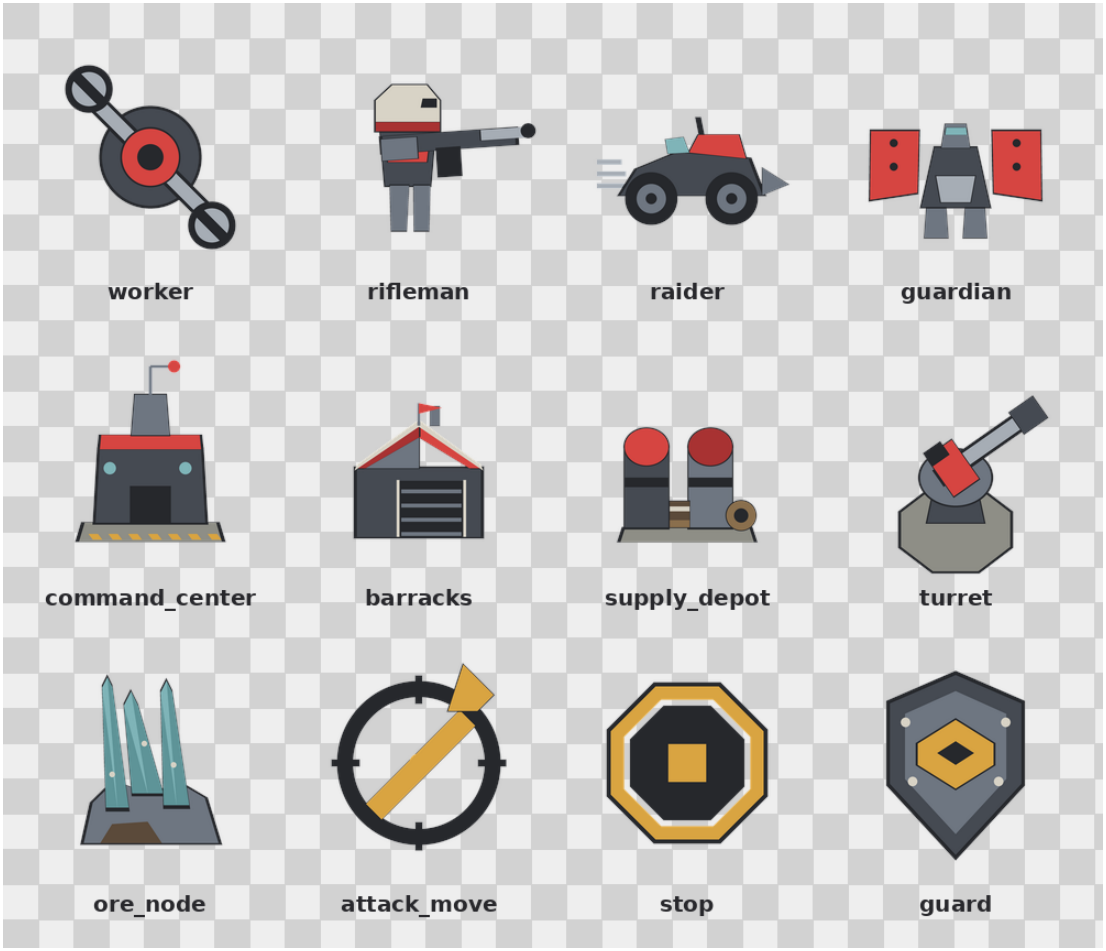


Figure 9-1: the D-001 HUD icon set on a checkerboard proving transparency.